

Configuration, Rules Taxonomy & Documentation

Project: T&C; Red Flag Scanner | Team: CorruptX

File: backend/risk_rules.json

Size: 20499 bytes

```

1 | {
2 |   "version": "1.0.0",
3 |   "taxonomy_name": "CorruptX CyberSecurity T&C Risk Taxonomy",
4 |   "categories": [
5 |     {
6 |       "id": "data_sharing",
7 |       "name": "Data Sharing",
8 |       "severity": "HIGH",
9 |       "weight": 15,
10 |      "color": "#ff3366",
11 |      "description": "Sharing, selling, transferring, or disclosing personal data to external third parties, business pa
12 |      "patterns": [
13 |        "share(?:s|d|ing)?\\s+(?:your|user|personal|sensitive)\\s+(?:information|data|details)\\s+with\\s+third\\s+parti
14 |        "disclose(?:s|d|ing)?\\s+(?:your|personal|user)\\s+(?:information|data)\\s+to\\s+(?:third\\s+parties|affiliates|
15 |        "sell(?:s|ing)?\\s+(?:or\\s+rent(?:s|ing)?\\s+)?(?:your|personal|user)\\s+(?:data|information)",
16 |        "transfer(?:s|red|ring)?\\s+(?:your|personal)\\s+data\\s+to\\s+(?:other\\s+entities|third\\s+parties|external)",
17 |        "commercial\\s+partners\\s+may\\s+(?:receive|access|use)\\s+your\\s+data",
18 |        "we\\s+reserve\\s+the\\s+right\\s+to\\s+share\\s+(?:aggregated|anonymized|de-identified|any)?\\s*(?:information|
19 |        "monetize(?:s|d|ing)?\\s+(?:user|personal)\\s+information",
20 |        "in\\s+the\\s+event\\s+of\\s+a\\s+merger(?:,|\\s+or)\\s+(?:acquisition|sale|bankruptcy).*data.*transferred"
21 |      ],
22 |      "keywords": [
23 |        "sell your data",
24 |        "share your information with third parties",
25 |        "disclose personal data to partners",
26 |        "monetize user information",
27 |        "share with advertisers"
28 |      ],
29 |      "explanation": "This clause allows the service to share, distribute, or sell your personal details with third-part
30 |      "why_it_matters": "Once your information is handed over to outside parties, you lose direct control over how it is
31 |      "recommendation": "Check if an opt-out mechanism is available or avoid submitting sensitive confidential data."
32 |    },
33 |    {
34 |      "id": "tracking",
35 |      "name": "Tracking & Telemetry",
36 |      "severity": "HIGH",
37 |      "weight": 14,
38 |      "color": "#ff3366",
39 |      "description": "Continuous device fingerprinting, behavioral tracking, cross-site telemetry, or keystroke monitori
40 |      "patterns": [
41 |        "track(?:s|ed|ing)?\\s+your\\s+(?:browsing|activity|behavior|location|online\\s+habits)",
42 |        "fingerprint(?:ing)?\\s+(?:your\\s+device|hardware|browser)",
43 |        "cross-site\\s+tracking",
44 |        "canvas\\s+fingerprinting",
45 |        "beacon(?:s)?|pixel\\s+tags?|web\\s+bugs?",
46 |        "record(?:s|ing)?\\s+(?:your\\s+screen|keystrokes|mouse\\s+movements|session\\s+replay)",
47 |        "monitor(?:s|ing)?\\s+(?:all|your)?\\s*(?:communications|browsing\\s+history|actions)",
48 |        "collect(?:s|ed|ing)?\\s+information\\s+about\\s+your\\s+activity\\s+across\\s+(?:third-party|other)\\s+websites
49 |      ],
50 |      "keywords": [
51 |        "cross-site tracking",
52 |        "device fingerprinting",
53 |        "session replay",
54 |        "keystroke recording",
55 |        "track your browsing activity"
56 |      ],
57 |      "explanation": "This clause enables active monitoring, device fingerprinting, or tracking of your online behavior
58 |      "why_it_matters": "Cross-site and hardware telemetry can build a persistent digital shadow profile of your habits,
59 |      "recommendation": "Use privacy-focused browsers, anti-fingerprinting shields, and strict tracker-blocking extensio
60 |    },

```

```

61 | {
62 |     "id": "data_collection",
63 |     "name": "Data Collection",
64 |     "severity": "HIGH",
65 |     "weight": 12,
66 |     "color": "#ff3366",
67 |     "description": "Excessive collection of sensitive personal data, biometrics, contacts, communications, or device i
68 |     "patterns": [
69 |         "collect(?:s|ed|ing)?\\s+(?:biometric|health|financial|genetic|sensitive)\\s+(?:information|data)",
70 |         "access(?:es)?\\s+(?:to\\s+)?your\\s+(?:contacts|address\\s+book|microphone|camera|stored\\s+files|photos)",
71 |         "scrape(?:s|d|ing)?\\s+(?:data|information|content)",
72 |         "collect(?:s|ed|ing)?\\s+(?:all|any)\\s+information\\s+you\\s+(?:transmit|submit|upload|post|create)",
73 |         "gather(?:s|ed|ing)?\\s+(?:details|information)\\s+from\\s+external\\s+public\\s+and\\s+private\\s+records",
74 |         "including\\s+but\\s+not\\s+limited\\s+to\\s+social\\s+security|credit\\s+card|bank\\s+account"
75 |     ],
76 |     "keywords": [
77 |         "biometric data",
78 |         "access to your contacts",
79 |         "microphone and camera access",
80 |         "sensitive personal data",
81 |         "financial information"
82 |     ],
83 |     "explanation": "The provider reserves extensive rights to gather invasive categories of personal data, including c
84 |     "why_it_matters": "Over-collection expands your attack surface; if the provider suffers a data breach, your sensit
85 |     "recommendation": "Review permissions requested on your device and restrict sensor/contact access."
86 | },
87 | {
88 |     "id": "arbitration",
89 |     "name": "Mandatory Arbitration",
90 |     "severity": "HIGH",
91 |     "weight": 14,
92 |     "color": "#ff3366",
93 |     "description": "Forced binding arbitration and waiver of court trials and collective class action lawsuits.",
94 |     "patterns": [
95 |         "binding\\s+arbitration",
96 |         "waive(?:r|s|d)?\\s+(?:any\\s+right\\s+to\\s+)?(?:a\\s+)?jury\\s+trial",
97 |         "waive(?:r|s|d)?\\s+(?:any\\s+right\\s+to\\s+participate\\s+in\\s+a\\s+)?class\\s+action",
98 |         "class\\s+action\\s+waiver",
99 |         "disputes\\s+shall\\s+be\\s+resolved\\s+(?:exclusively\\s+)?(?:by|through)\\s+(?:confidential|individual|binding
100 |         "no\\s+class\\s+or\\s+representative\\s+actions?",
101 |         "agree\\s+not\\s+to\\s+sue\\s+in\\s+court"
102 |     ],
103 |     "keywords": [
104 |         "binding arbitration",
105 |         "class action waiver",
106 |         "waive jury trial",
107 |         "resolve disputes through arbitration",
108 |         "no class action"
109 |     ],
110 |     "explanation": "You surrender your constitutional right to take this company to court or participate in a joint co
111 |     "why_it_matters": "Arbitration proceedings are usually private, costly for individuals, and systematically favor c
112 |     "recommendation": "Check if an arbitration opt-out letter can be submitted within 30 days of registration."
113 | },
114 | {
115 |     "id": "unilateral_changes",
116 |     "name": "Unilateral Changes",
117 |     "severity": "HIGH",
118 |     "weight": 13,
119 |     "color": "#ff3366",
120 |     "description": "Changing legal terms or privacy policies at any time without notifying users or requiring fresh co
121 |     "patterns": [
122 |         "reserve\\s+the\\s+right\\s+to\\s+(?:modify|change|alter|update)\\s+(?:these\\s+terms|this\\s+agreement|policy)\\s+",
123 |         "(?:modify|change|update)\\s+without\\s+(?:prior\\s+)?notice",
124 |         "continued\\s+use\\s+(?:of\\s+the\\s+service\\s+)?(?:constitutes|shall\\s+constitute|signifies)\\s+(?:your\\s+)?",
125 |         "without\\s+obligation\\s+to\\s+notify\\s+you",
126 |         "effective\\s+immediately\\s+upon\\s+posting"
127 |     ],
128 |     "keywords": [
129 |         "modify without prior notice",

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130 |         "at our sole discretion without notice",
131 |         "continued use constitutes acceptance",
132 |         "change these terms at any time"
133 |     ],
134 |     "explanation": "The company can alter policies, fees, or data handling rules at any time, and your continued usage",
135 |     "why_it_matters": "You could wake up to new invasive terms or higher subscription fees without ever receiving an e",
136 |     "recommendation": "Periodically review terms updates or monitor policy changes using change-detection tools."
137 | },
138 | {
139 |     "id": "liability_waiver",
140 |     "name": "Liability Waiver",
141 |     "severity": "MEDIUM",
142 |     "weight": 9,
143 |     "color": "#ffb700",
144 |     "description": "Total disclaimer of liability for data loss, service outages, bugs, or security vulnerabilities.",
145 |     "patterns": [
146 |         "provided\\s+[\\']?as\\s+is[\\']?\\s+and\\s+[\\']?as\\s+available[\\']?",
147 |         "disclaim(?:s)?\\s+(?:all|any)\\s+warranties",
148 |         "(?:under\\s+no\\s+circumstances\\s+shall|in\\s+no\\s+event\\s+shall)\\s+(?:the\\s+company|we|us|our\\s+affiliat",
149 |         "limit(?:ation)?\\s+of\\s+liability\\s+shall\\s+not\\s+exceed\\s+(?:the\\s+amount\\s+paid|\\$\\d+|one\\s+hundred",
150 |         "not\\s+liable\\s+for\\s+(?:any\\s+)?(?:data\\s+loss|security\\s+breaches|unauthorized\\s+access)"
151 |     ],
152 |     "keywords": [
153 |         "provided as is",
154 |         "in no event shall we be liable",
155 |         "disclaim all warranties",
156 |         "not liable for data loss or breach",
157 |         "limitation of liability"
158 |     ],
159 |     "explanation": "The company provides the product 'as-is' and completely disclaims legal responsibility if their so",
160 |     "why_it_matters": "If a critical security bug causes data corruption or identity theft, this clause shields the ve",
161 |     "recommendation": "Never rely solely on this vendor for sole data backups; maintain external independent copies."
162 | },
163 | {
164 |     "id": "auto_renewal",
165 |     "name": "Auto-Renewal & Recurring Billing",
166 |     "severity": "MEDIUM",
167 |     "weight": 8,
168 |     "color": "#ffb700",
169 |     "description": "Automatic recurring subscription renewals, often accompanied by strict cancellation timelines.",
170 |     "patterns": [
171 |         "auto(?:matically)?-?renew(?:s|ed|ing)?",
172 |         "renew(?:s|ed|ing)?\\s+automatically\\s+unless\\s+cancelled",
173 |         "recurring\\s+(?:billing|charge|subscription|fee)",
174 |         "charge(?:s|d)?\\s+your\\s+(?:payment\\s+method|credit\\s+card)\\s+without\\s+(?:additional|further)\\s+authoriz",
175 |         "cancel(?:ation)?\\s+at\\s+least\\s+\\d+\\s+days\\s+prior\\s+to\\s+(?:the\\s+end|renewal)",
176 |         "non-refundable\\s+(?:fees|charges|payments)"
177 |     ],
178 |     "keywords": [
179 |         "automatically renew",
180 |         "recurring subscription",
181 |         "cancel at least 24 hours prior",
182 |         "non-refundable fees",
183 |         "charge without further authorization"
184 |     ],
185 |     "explanation": "Your subscription will silently renew and charge your card automatically unless manually terminate",
186 |     "why_it_matters": "Users frequently get trapped in zombie billing cycles for unused services due to obscure cancel",
187 |     "recommendation": "Set calendar alerts 3-5 days ahead of billing cycles or use virtual credit cards with spend lim",
188 | },
189 | {
190 |     "id": "account_termination",
191 |     "name": "Account Termination",
192 |     "severity": "MEDIUM",
193 |     "weight": 8,
194 |     "color": "#ffb700",
195 |     "description": "Suspension or deletion of user accounts and content without prior notice, justification, or refund",
196 |     "patterns": [
197 |         "terminate\\s+(?:or\\s+suspend\\s+)?(?:your\\s+account|access)\\s+at\\s+(?:any\\s+time|our\\s+sole\\s+discretion",
198 |         "(?:terminate|suspend)\\s+without\\s+(?:prior\\s+)?notice\\s+(?:or\\s+liability)?",

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199 |         "for\\s+any\\s+reason\\s+or\\s+(?:for\\s+)?no\\s+reason",
200 |         "forfeit(?:ure)?\\s+(?:of\\s+)?(?:all|any)\\s+(?:content|credits|funds|data)\\s+upon\\s+termination",
201 |         "no\\s+obligation\\s+to\\s+refund\\s+(?:unused\\s+)?fees"
202 |     ],
203 |     "keywords": [
204 |         "terminate at our sole discretion",
205 |         "suspend without notice",
206 |         "for any reason or no reason",
207 |         "forfeit content upon termination"
208 |     ],
209 |     "explanation": "The provider can ban your account, delete your assets, and block your access at will without explanation.",
210 |     "why_it_matters": "You risk sudden loss of work, purchases, access to historical records, or communication channels.",
211 |     "recommendation": "Regularly export and download backups of your cloud files, projects, and transactions."
212 | },
213 | {
214 |     "id": "third_party_access",
215 |     "name": "Third-Party Access",
216 |     "severity": "HIGH",
217 |     "weight": 11,
218 |     "color": "#ff3366",
219 |     "description": "Permitting outside integrations, ad networks, or vendors to access user accounts and content.",
220 |     "patterns": [
221 |         "third-party\\s+(?:service\\s+providers|vendors|contractors|integrations)\\s+may\\s+access",
222 |         "not\\s+responsible\\s+for\\s+(?:the\\s+privacy\\s+practices|actions|content)\\s+of\\s+third\\s+parties",
223 |         "allow(?:s)?\\s+third\\s+parties\\s+to\\s+(?:serve\\s+ads|collect\\s+data|place\\s+cookies)",
224 |         "integrated\\s+third-party\\s+services\\s+governed\\s+by\\s+their\\s+own\\s+terms",
225 |         "share\\s+with\\s+sub-processors\\s+without\\s+notification"
226 |     ],
227 |     "keywords": [
228 |         "third-party service providers may access",
229 |         "not responsible for third-party privacy",
230 |         "third parties to serve ads and collect",
231 |         "governed by third-party policies"
232 |     ],
233 |     "explanation": "External software vendors or third-party plug-ins can interface directly with your account or read your data.",
234 |     "why_it_matters": "Third-party supply chain vulnerabilities are a leading vector for corporate data leaks and identity theft.",
235 |     "recommendation": "Audit and disconnect unused connected apps and third-party account authorizations."
236 | },
237 | {
238 |     "id": "data_retention",
239 |     "name": "Indefinite Data Retention",
240 |     "severity": "MEDIUM",
241 |     "weight": 7,
242 |     "color": "#ffb700",
243 |     "description": "Retaining personal data, messages, or backups indefinitely even after account deletion.",
244 |     "patterns": [
245 |         "retain(?:s|ed|ing)?\\s+(?:your\\s+)?(?:data|information|content)\\s+(?:indefinitely|for\\s+an\\s+unlimited\\s+period)",
246 |         "may\\s+retain\\s+(?:archived|backup|residual)\\s+copies\\s+even\\s+after\\s+(?:you\\s+delete|account\\s+closure)",
247 |         "no\\s+obligation\\s+to\\s+delete\\s+(?:your\\s+)?(?:information|data)",
248 |         "retain\\s+and\\s+use\\s+your\\s+information\\s+to\\s+comply\\s+with\\s+legal\\s+obligations.*resolve\\s+disputes"
249 |     ],
250 |     "keywords": [
251 |         "retain data indefinitely",
252 |         "retain backup copies even after deletion",
253 |         "as long as we deem necessary",
254 |         "no obligation to delete"
255 |     ],
256 |     "explanation": "Deleting your account does not guarantee deletion of your personal records, which may be archived or used for legal purposes.",
257 |     "why_it_matters": "Zombie data retained forever creates a permanent target for future database breaches or subpoenas.",
258 |     "recommendation": "Exercise formal GDPR/CCPA 'Right to Erasure' requests where legally applicable."
259 | },
260 | {
261 |     "id": "location_tracking",
262 |     "name": "Location Tracking",
263 |     "severity": "HIGH",
264 |     "weight": 12,
265 |     "color": "#ff3366",
266 |     "description": "Harvesting precise GPS coordinates, Wi-Fi triangulation, cell tower IDs, or continuous background location data.",
267 |     "patterns": [

```

```

268 |         "collect(?:s|ed|ing)?\\s+(?:precise|exact|real-time|continuous)?\\s*(?:location|geolocation|gps)\\s+data",
269 |         "location\\s+(?:even\\s+when\\s+the\\s+app\\s+is\\s+closed|in\\s+the\\s+background)",
270 |         "track(?:s|ed|ing)?\\s+(?:your\\s+physical\\s+location|whereabouts|movements)",
271 |         "derive\\s+location\\s+from\\s+(?:ip\\s+address|wi-fi\\s+access\\s+points|cell\\s+towers|bluetooth)"
272 |     ],
273 |     "keywords": [
274 |         "precise geolocation data",
275 |         "location in the background",
276 |         "track your physical location",
277 |         "real-time location tracking"
278 |     ],
279 |     "explanation": "The app tracks or approximates your real-world geographic coordinates, potentially even when running in the background.",
280 |     "why_it_matters": "Real-time location records expose your residence, workplace, daily routines, and intimate personal information.",
281 |     "recommendation": "Set system location permissions to 'Only While Using App' or disable location entirely if not needed."
282 | },
283 | {
284 |     "id": "advertising_profiling",
285 |     "name": "Advertising & Profiling",
286 |     "severity": "MEDIUM",
287 |     "weight": 8,
288 |     "color": "#ffb700",
289 |     "description": "Targeted behavioral ads, algorithmic psychological profiling, and interest-based segmentation.",
290 |     "patterns": [
291 |         "personalized\\s+(?:advertisements|ads|marketing|promotions)",
292 |         "interest-based\\s+advertising",
293 |         "create\\s+(?:a\\s+)?profile\\s+about\\s+you",
294 |         "target(?:ed)?\\s+advertising\\s+based\\s+on\\s+your\\s+(?:preferences|activity|browsing)",
295 |         "deliver\\s+relevant\\s+ads\\s+by\\s+analyzing\\s+your\\s+(?:interests|profile|demographics)"
296 |     ],
297 |     "keywords": [
298 |         "personalized advertisements",
299 |         "interest-based advertising",
300 |         "create a profile about you",
301 |         "targeted advertising based on"
302 |     ],
303 |     "explanation": "Your activity, searches, and demographic attributes are used to assemble an advertising profile for targeted marketing.",
304 |     "why_it_matters": "Targeted profiling can exploit behavioral vulnerabilities, shape opinions, and expose your private information.",
305 |     "recommendation": "Opt out of personalized advertising within service settings and reset your advertising ID."
306 | },
307 | {
308 |     "id": "user_rights",
309 |     "name": "User Rights Surrender",
310 |     "severity": "HIGH",
311 |     "weight": 11,
312 |     "color": "#ff3366",
313 |     "description": "Surrendering intellectual property rights, granting irrevocable global content licenses, or moral rights waiver.",
314 |     "patterns": [
315 |         "grant(?:s)?\\s+(?:us\\s+)?(?:a\\s+)?(?:worldwide|perpetual|irrevocable|royalty-free|unrestricted)\\s+(?:license|agreement)",
316 |         "waive(?:s|d)?\\s+(?:all\\s+)?moral\\s+rights",
317 |         "you\\s+assign\\s+(?:all\\s+)?(?:ownership|intellectual\\s+property|rights)\\s+to\\s+us",
318 |         "we\\s+own\\s+all\\s+(?:feedback|submissions|content|ideas)\\s+provided\\s+by\\s+you",
319 |         "right\\s+to\\s+sublicense\\s+and\\s+commercialize\\s+your\\s+content"
320 |     ],
321 |     "keywords": [
322 |         "perpetual irrevocable royalty-free license",
323 |         "worldwide license to use and modify",
324 |         "waive moral rights",
325 |         "assign all intellectual property",
326 |         "sublicense and commercialize your content"
327 |     ],
328 |     "explanation": "You grant the company a permanent, global, royalty-free license to use, publish, monetize, or adapt your content.",
329 |     "why_it_matters": "You may lose commercial ownership or credit for creative works, files, or original designs uploaded to the platform.",
330 |     "recommendation": "Avoid storing or creating proprietary work or copyrighted media on platforms demanding broad IP licenses."
331 | },
332 | {
333 |     "id": "cancellation_restrictions",
334 |     "name": "Cancellation Restrictions",
335 |     "severity": "LOW",
336 |     "weight": 5,

```

```
337 |         "color": "#38bdf8",
338 |         "description": "Complex hurdle-filled cancellation procedures, early termination penalties, or restricted contact
339 |         "patterns": [
340 |             "cancel(?:lation)?\\s+(?:must\\s+be\\s+made\\s+by|only\\s+by)\\s+(?:phone|written\\s+notice|certified\\s+mail)",
341 |             "early\\s+termination\\s+fee(?:s)?",
342 |             "penalt(?:y|ies)\\s+for\\s+(?:early\\s+)?cancellation",
343 |             "must\\s+provide\\s+(?:at\\s+least\\s+)?(?:30|60|90)\\s+days\\s+(?:prior\\s+)?notice\\s+to\\s+cancel",
344 |             "cannot\\s+be\\s+cancelled\\s+online"
345 |         ],
346 |         "keywords": [
347 |             "early termination fee",
348 |             "cancel only by phone",
349 |             "penalty for cancellation",
350 |             "certified mail to cancel"
351 |         ],
352 |         "explanation": "The provider imposes restrictive hurdles or fees to make terminating your account or contract deli
353 |         "why_it_matters": "Dark pattern cancellation flows prevent easy departure, forcing unneeded payments or unnecessar
354 |         "recommendation": "Check the exact cancellation procedures prior to providing payment details or signing long agre
355 |     }
356 | ]
357 | }
```

File: backend/sample_policies.json

Size: 7003 bytes

```
1 | {
2 |   "samples": [
3 |     {
4 |       "id": "high_risk_social",
5 |       "title": "HyperNet Social App Terms & Privacy Policy",
6 |       "expected_risk": "HIGH RISK",
7 |       "description": "A consumer mobile application packed with invasive tracking, cross-site telemetry, binding arbitra
8 |       "content": "Welcome to HyperNet. By accessing or using our services, mobile application, or websites, you agree to
9 |     },
10 |    {
11 |      "id": "moderate_risk_saas",
12 |      "title": "NovaDesk Cloud Workspace Agreement",
13 |      "expected_risk": "CAUTION",
14 |      "description": "A business SaaS platform with commercial auto-renewal, standard liability caps, unilateral mainter
15 |      "content": "NovaDesk Cloud Services Master Subscription Agreement.\n\n1. SCOPE AND SERVICES\nNovaDesk provides onl
16 |    },
17 |    {
18 |      "id": "safe_open_source",
19 |      "title": "AegisGuard Privacy-First Client Terms",
20 |      "expected_risk": "SAFE",
21 |      "description": "An open-source, local-first utility with end-to-end encryption, zero tracking, transparent cancell
22 |      "content": "AegisGuard Terms of Use and Privacy Charter.\n\n1. ZERO KNOWLEDGE ARCHITECTURE\nAegisGuard operates as
23 |    }
24 |  ]
25 | }
```

File: backend/requirements.txt

Size: 54 bytes

```
1 | Flask>=3.0.0
2 | flask-cors>=4.0.0
3 | beautifulsoup4>=4.12.0
```

File: README.md

Size: 11291 bytes

```

1 | # T&C RED FLAG SCANNER
2 | &gt; ***"Before you click Agree, know what you're agreeing to."**
3 | &gt; **Team: CorruptX** | Cybersecurity Hackathon Project
4 |
5 | ---
6 |
7 | ## 1. Project Overview
8 |
9 | **T&C Red Flag Scanner** is an AI-powered cybersecurity application designed to analyze Terms & Conditions, Privacy Policies, and End User License Agreements (EULAs).
10 |
11 | Built with a **100% offline-first architecture**, the application guarantees zero third-party cloud data leakage—meaning all analysis happens locally on your device.
12 |
13 | ---
14 |
15 | ## 2. Key Features
16 |
17 | - **Automated Clause Segmentation**: Splits complex legal prose into discrete clauses while preserving decimal sections.
18 | - **14-Category Risk Taxonomy**: Detects patterns across critical vectors including:
19 |   1. Data Sharing & Monetization
20 |   2. Tracking & Hardware Telemetry (Fingerprinting, Session Replay)
21 |   3. Excessive Data Collection (Sensors, Contacts, Biometrics)
22 |   4. Mandatory Binding Arbitration & Class Action Waivers
23 |   5. Unilateral Policy Modifications without Notice
24 |   6. Broad Liability Waivers & Warranty Disclaimers
25 |   7. Auto-Renewal & Recurring Billing Traps
26 |   8. Unilateral Account Termination & Content Forfeiture
27 |   9. Third-Party Vendor Access
28 |   10. Indefinite Data Retention
29 |   11. Real-Time Location & Background Tracking
30 |   12. Behavioral Advertising & Demographic Profiling
31 |   13. Content Licensing & User Rights Surrender
32 |   14. Restrictive Cancellation Penalties
33 | - **Negation & Privacy Protection Detection**: Intelligently distinguishes between risky clauses and positive privacy protections.
34 | - **Deterministic 0-100 Risk Score**: Computes a mathematically grounded exposure score and assigns clear risk tiers:
35 |   - `SAFE` (0-30)
36 |   - `CAUTION` (31-60)
37 |   - `HIGH RISK` (61-100)
38 | - **Interactive Policy Highlighting & Inspector**: Renders full document text with inline color-coded risk spans (`.highlight` classes).
39 | - **Client-Side Offline Engine**: Bundled with a dual-execution architecture. If the Python backend is running, it uses the local engine; otherwise, it falls back to a browser-based JavaScript engine.
40 | - **Offline History & Reporting**: Stores previous scans locally using browser `localStorage` with view, delete, and export capabilities.
41 |
42 | ---
43 |
44 | ## 3. Technology Stack
45 |
46 | - **Backend**: Python 3 (Flask, Flask-CORS, BeautifulSoup4)
47 | - **Analysis Engine**: Rule-based NLP pattern matcher and weighted scoring matrix (`analyzer.py` + `risk_rules.json`)
48 | - **Frontend**: Modern Vanilla HTML5, CSS3 (Cyberpunk/Dark Cybersecurity Theme, Glassmorphism, CSS Grid), Vanilla ES6+ JavaScript
49 | - **Icons**: Local inline SVGs (zero CDN dependency)
50 | - **Storage**: Client-side `localStorage` and `sessionStorage`
51 |
52 | ---
53 |
54 | ## 4. Folder Structure
55 |
56 | ```
57 | tc-red-flag-scanner/
58 | └─
59 |   └─ backend/
60 |       └─ app.py           # Flask API server & static router
61 |       └─ analyzer.py      # Core NLP rule engine & risk scorer
62 |       └─ risk_rules.json  # Cybersecurity risk taxonomy (14 categories)
63 |       └─ sample_policies.json # Preloaded hackathon demo policies
64 |       └─ test_analyzer.py  # Automated test verification script
65 |       └─ requirements.txt  # Python dependencies
66 |   └─ frontend/
67 |

```

```

68 | ■ ■■■ index.html           # Landing page with hero, mockup & team info
69 | ■ ■■■ scanner.html        # Main analysis console & scanning modal
70 | ■ ■■■ results.html        # Results dashboard with circular risk meter
71 | ■ ■■■ history.html         # Scan history management
72 | ■ ■■■ about.html          # Mission, cybersecurity context & team details
73 | ■
74 | ■ ■■■ css/
75 | ■ ■ ■■■ style.css         # Global theme, design system & typography
76 | ■ ■ ■■■ scanner.css       # Textarea, dropzone, and radar scanning overlay
77 | ■ ■ ■■■ results.css       # Circular gauge, metrics, cards & highlighter
78 | ■ ■ ■■■ responsive.css    # Mobile, tablet, and desktop breakpoints
79 | ■
80 | ■ ■■■ js/
81 | ■ ■■■ rules_data.js       # Offline bundled taxonomy
82 | ■ ■■■ sample_data.js      # Offline bundled demo policies
83 | ■ ■■■ client_analyzer.js  # Standalone client-side NLP engine
84 | ■ ■■■ app.js              # Shared state, notifications & localStorage
85 | ■ ■■■ scanner.js          # Scanner page controller & animation
86 | ■ ■■■ results.js          # Circular meter, highlighter & inspector
87 | ■ ■■■ history.js          # History manager & report exporter
88 | ■
89 | ■■■ README.md             # Complete technical documentation
90 | ■■■ .gitignore            # Git ignore rules
91 | ---
92 |
93 | ---
94 |
95 | ## 5. Installation & Setup
96 |
97 | ### Prerequisites
98 | - Python 3.9+ (Python 3.10, 3.11, 3.12, 3.14 supported)
99 | - Any modern web browser (Chrome, Edge, Firefox, Brave, Safari)
100 |
101 | ### Option A: Running with Local Flask Server (Recommended)
102 |
103 | 1. Open PowerShell / Terminal in the `tc-red-flag-scanner/backend` directory:
104 |    ```bash
105 |    cd "backend"
106 |    ```
107 |
108 | 2. Install dependencies:
109 |    ```bash
110 |    pip install -r requirements.txt
111 |    ```
112 |
113 | 3. Start the Flask application:
114 |    ```bash
115 |    python app.py
116 |    ```
117 |
118 | 4. Open your browser and navigate to:
119 |    ```
120 |    http://127.0.0.1:5000
121 |    ```
122 |
123 | ### Option B: Running 100% Offline Standalone (Zero Server Required)
124 |
125 | You can run the entire application completely without a web server or Python:
126 | 1. Navigate to the `frontend/` folder.
127 | 2. Double-click `index.html` or open it directly in any browser (`file:///.../frontend/index.html`).
128 | 3. The built-in client-side analyzer (`client_analyzer.js`) automatically activates and executes scans locally in your browser.
129 |
130 | ---
131 |
132 | ## 6. REST API Endpoints
133 |
134 | | Method | Endpoint | Description |
135 | |---|---|---|
136 | | `GET` | `/api/health` | Service health status and active rule categories count |

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137 | | `GET` | `/api/rules` | Complete active risk rules taxonomy (`risk_rules.json`) |
138 | | `GET` | `/api/demo` | Curated sample policies for instant hackathon evaluation |
139 | | `POST` | `/api/scan` | Main analysis endpoint. Accepts JSON or multipart file uploads |
140 |
141 | ### Example Scan Request:
142 | ```bash
143 | curl -X POST http://127.0.0.1:5000/api/scan \
144 |   -H "Content-Type: application/json" \
145 |   -d '{"text": "We reserve the right to share your personal information with third parties. All disputes shall be resolved in arbitration."}'
146 | ```
147 |
148 | ### Example Response:
149 | ```json
150 | {
151 |   "document_name": "Test Contract",
152 |   "analyzed_at": "2026-09-22T23:15:30.123456",
153 |   "risk_score": 68,
154 |   "risk_level": "HIGH RISK",
155 |   "clauses_analyzed": 2,
156 |   "red_flags": 2,
157 |   "severity_counts": {
158 |     "HIGH": 2,
159 |     "MEDIUM": 0,
160 |     "LOW": 0
161 |   },
162 |   "categories": [
163 |     { "category_id": "data_sharing", "category_name": "Data Sharing", "severity": "HIGH", "count": 1 },
164 |     { "category_id": "arbitration", "category_name": "Mandatory Arbitration", "severity": "HIGH", "count": 1 }
165 |   ],
166 |   "flagged_clauses": [
167 |     {
168 |       "index": 0,
169 |       "text": "We reserve the right to share your personal information with third parties.",
170 |       "category_name": "Data Sharing",
171 |       "severity": "HIGH",
172 |       "explanation": "This clause allows the service to share, distribute, or sell your personal details with third-parties.",
173 |       "why_it_matters": "Once your information is handed over to outside parties, you lose direct control...",
174 |       "recommendation": "Check if an opt-out mechanism is available or avoid submitting sensitive confidential data."
175 |     }
176 |   ]
177 | }
178 | ```
179 |
180 | ---
181 |
182 | ## 7. Risk Scoring Methodology
183 |
184 | The risk score is calculated using a deterministic algorithm:
185 | 1. Clause Weighting:
186 |   - `HIGH` severity clauses: 16 points each
187 |   - `MEDIUM` severity clauses: 8 points each
188 |   - `LOW` severity clauses: 4 points each
189 | 2. Exposure Density Factor:
190 |   - Compares the ratio of flagged clauses to total document clauses:
191 |     
$$\text{Density Bonus} = \min\left(20, \left\lfloor \frac{\text{Flagged Clauses}}{\text{Total Clauses}} \right\rfloor \times 35\right)$$

192 | 3. Composite Normalization:
193 |     
$$\text{Score} = \min\left(100, \left(\text{Weighted Sum} \times 0.82\right) + \text{Density Bonus}\right)$$

194 | 4. Safety Threshold Guardrails:
195 |   - If  $\geq 3$  `HIGH` severity red flags are detected, the minimum score floor is bounded to $68$ (`HIGH RISK`).
196 |   - If  $\geq 1$  `HIGH` severity red flag is detected, score floor is bounded to $42$ (`CAUTION`).
197 |
198 | ---
199 |
200 | ## 8. Hackathon Demo Instructions
201 |
202 | 1. Launch the application (`python backend/app.py` or open `frontend/index.html`).
203 | 2. Click "Scan a Document" or navigate to `/scanner`.
204 | 3. Click the "Load Demo Policy" button in the toolbar. This loads the realistic "HyperNet Social App Terms", featuring various red flags.
205 | 4. Click "Start Security Scan".

```

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206 | 5. Observe the 4-step animated scanning overlay (`Extracting clauses... -&gt; Classifying risks... -&gt; Calculating ris
207 | 6. View the Results Dashboard:
208 |     - Inspect the circular risk score gauge (~100/100 HIGH RISK).
209 |     - Review detected red flag cards with plain-English translations and recommendations.
210 |     - Switch to the "Interactive Policy Analysis" tab and click on highlighted spans to test the real-time Clause I
211 |     - Check the "Risk Breakdown" tab for category rankings.
212 | 7. Click "History" in the top navigation to verify persistent local logging.
213 |
214 | ---
215 |
216 | ## 9. Future AI Architecture & Roadmap
217 |
218 | ```
219 | Document / Policy Input
220 |     ↓
221 | Text Normalization & Cleaning
222 |     ↓
223 | Intelligent Clause Segmentation
224 |     ↓
225 | LLM / NLP Embedding & Zero-Shot Classification
226 | (e.g., Local Llama 3 / Gemma 2B via ONNX Runtime / Ollama)
227 |     ↓
228 | Cybersecurity Risk Taxonomy Alignment
229 |     ↓
230 | Plain-English Simplification & Defense Synthesis
231 |     ↓
232 | Interactive Results Dashboard
233 | ```
234 |
235 | The modular design of `ClauseAnalyzer` ensures that swapping regex rules with a HuggingFace Transformer pipeline or loca
236 |
237 | ---
238 |
239 | ## 10. Important Product Disclaimer
240 |
241 | &gt; Disclaimer: This tool provides automated risk analysis for informational and educational cybersecurity purposes
242 |
243 | ---
244 |
245 | ## 11. Team Information
246 |
247 | Team CorruptX
248 | - Vyom Khanna — Team Lead (Architecture & Full-Stack Engineering)
249 | - Nalin Shakher — Team Member (NLP Engine, Risk Taxonomy & Offline Pipeline)
250 | - Aayush Verma — Team Member (Cybersecurity UI/UX & Interactive Clause Inspector)

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